RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 15th April 2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: FACILITIES PLANNING AND DESIGN		**Course Code : IM365TDA**

Note:
1. Answer all the Questions.

PART – A

SI. No.	Questions	M	BT	CO
1	State any two factors influencing plant location.	02	L2	CO1
2	Identify the suitable layout for a hospital and justify your choice.	02	L2	CO1
3	Which layout is suitable for job-order production? Give reason.	02	L1	CO2
4	Analyze any two causes of inadequate facilities planning with examples.	02	L1	CO1
5	Propose a basic facilities planning strategy for a new manufacturing firm.	02	L2	CO2

PART – B

SI. No.	Questions	M	BT	CO
1.	Explain the concept of facilities planning and discuss its significance in modern industries.	10	L1	CO1
2.	Differentiate between facilities planning process and strategic planning process in terms of meaning, scope, objective, time horizon and focus.	10	L2	CO2
3.	Explain the factors influencing plant location with suitable examples.	10	L3	CO3
4.	Recommend a suitable plant layout for a manufacturing firm and justify your choice with merits and demerits.	10	L4	CO4
5.	Potential locations A, B and C have the cost structure as shown below for producing a product expected to sell for Rs. 4250 per unit. Site | Fixed Cost/year (Rs) | Variable Cost/unit (Rs) A | 85,00,000 | 775 B | 65,00,000 | 2225 C | 70,00,000 | 1460 Find the most economical location for an expected volume of 6250 units per year.	10	L2	CO2

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Date : 16th April 2026	**CIE - 1**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Service Operations Management		**Course Code : IM365TDB**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is the primary goal of Service Operations Management?	2	1	1
2.	Name two major challenges faced by service operations managers in today's dynamic environment.	2	2	2
3.	Identify any 4 tools that have been used on you to collect information to enhance customer satisfaction.	2	3	1
4.	What type of service process is typically used in a fast-food restaurant, and why?	2	3	2
5.	How does service operations management differ from manufacturing operations management?	2	2	2
	Part – B			
1	What are the different types of service processes? Cite the example of any service organization and highlight the different processes involved in that organization.	10	2	2
2	Apply the concept of the "Poka-Yoke" (mistake-proofing) methodology to a service setting of your choice. Explain how you would design both warning and control types of poka-yokes to manage the operational risks associated with customer variability and employee variability.	10	3	2
3	Explain "Service concept" using the tabular column as applied to a "New age engineering education institute".	10	5	3
4	You are tasked with launching a new, subscription-based "tele-health" service that connects patients with doctors via video calls 24/7. This service is highly intangible and faces significant variability in demand. Create a set of success metrics that specifically measure the "moments of truth" in the virtual interaction, addressing the challenge of measuring quality for an intangible product. Hint: Go up to 3 Moments of Truth starting from Zeroth. Use tabular column and avoid writing in paragraphs.	10	4	2
5	You are a Service Operations Manager of an online retail store. What challenges would you may face as the manager and what measures would you take to overcome them?	10	3	3

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Date : 15th April 2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Design of Experiments		**Course Code : IM365TDD**

Note:
1. Answer all the Questions.
2. Use of statistical tables permitted

Sl. No	Questions	M	BT	CO
	Part – A			
1.	List two applications of design of experiments.	1	1	CO1
2.	What are the two advantages of replication?	2	1	CO1
3.	_____ is an example of larger-the-better type quality characteristic.	1	1	CO2
4.	According to Genichi Taguchi "Quality is the _____ a product causes to the society"	2	2	CO2
5.	Identify response, controllable and noise factors for gas welding process.	2	2	CO1
6.	What are the steps in the engineering design problem strategy?	2	2	CO2
	Part – B			
1	Enumerate the guidelines for the statistical approach in designing and analyzing an experiment.	10	2	CO1
2	Trace the history of statistical design. Elucidate the important stages.	10	3	CO1
3	Write the expressions for quality loss function of various quality characteristics. Illustrate with sketches. List the situations in which these quality characteristics are applicable.	10	2	CO2
4	Identify the quality control activities during various product realization steps. Comment on the ability of these activities to reduce effect of noise factors.	05	3	CO2

<table>
<tr>
<td>5</td>
<td>An industrial engineer employed by a beverage bottler is interested in the effects of two different types of 32-ounce bottles (A) on the time to deliver 12-bottle cases of the product. The two bottle types are glass and plastic. Two workers (B) are used to perform a task consisting of moving 40 cases of the product 50 feet on a standard type of hand truck and stacking the cases in a display. Four replicates of a 2^2 factorial design are performed, and the times observed are listed in the following table:
<table>
<tr><td rowspan="2">Treatment Combination</td><td colspan="4">Replicates</td></tr>
<tr><td>I</td><td>II</td><td>III</td><td>IV</td></tr>
<tr><td>1</td><td>5.12</td><td>4.89</td><td>4.98</td><td>5.00</td></tr>
<tr><td>a</td><td>6.65</td><td>6.24</td><td>5.49</td><td>5.55</td></tr>
<tr><td>b</td><td>4.95</td><td>4.43</td><td>4.27</td><td>4.25</td></tr>
<tr><td>ab</td><td>5.28</td><td>4.91</td><td>4.75</td><td>4.71</td></tr>
</table>
i. Estimate the factor effects.
ii. Prepare an analysis of variance table and determine which factors are important in explaining time to deliver.</td>
<td>15</td>
<td>2</td>
<td>CO4</td>
</tr>
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Date : 02nd May 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Facilities Planning and Design		**Course Code : IM365TDA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Facilities planning are subdivided into the subjects of facilities _____ and facilities _____.	(02)	L1	CO1
2.	State two objectives of facilities planning.	(02)	L2	CO1
3.	List two factors of facilities location.	(02)	L1	CO1
4.	Expand CRAFT and COFAD.	(02)	L1	CO1
5.	What is the basic principle of the CORELAP algorithm, and how is the first department selected?	(02)	L1	CO1
	Part – B			
1.	Enumerate and analyse the characteristics of all the facilities of a supply chain to achieve excellence.	10	L2	CO1
2.	Apply the Winning facilities planning process methodology for facilities planning of a manufacturing organization, and provide your analysis summary.	10	L3	CO3
3.	Perform a comparative assessment of the performance characteristics of process layout, product layout and cellular layout.	10	L2	CO2

<table>
<tr>
<td>4.</td>
<td>An Auto Parts warehouse has requested that a new layout be designed for their main warehouse. The warehouse has 8 major activity centers. The current building area can accommodate 10 X 24 unit squares. Develop the new layout using ALDEP using the data given below.
<table>
<tr><th>Code</th><th>Department Name</th><th>Area (m^2)</th></tr>
<tr><td>1</td><td>Storage</td><td>12800</td></tr>
<tr><td>2</td><td>Counter</td><td>8800</td></tr>
<tr><td>3</td><td>Paint Room</td><td>9200</td></tr>
<tr><td>4</td><td>Receiving</td><td>12000</td></tr>
<tr><td>5</td><td>Office</td><td>8800</td></tr>
<tr><td>6</td><td>Parts Bins</td><td>13200</td></tr>
<tr><td>7</td><td>Tailpipe Racks</td><td>13200</td></tr>
<tr><td>8</td><td>Restroom</td><td>8000</td></tr>
</table>
<table>
<tr><td>1</td><td colspan="7">——-</td></tr>
<tr><td>2</td><td>X</td><td colspan="6">——-</td></tr>
<tr><td>3</td><td>O</td><td>O</td><td colspan="5">——-</td></tr>
<tr><td>4</td><td>A</td><td>X</td><td>U</td><td colspan="4">——-</td></tr>
<tr><td>5</td><td>U</td><td>E</td><td>U</td><td>U</td><td colspan="3">——-</td></tr>
<tr><td>6</td><td>O</td><td>A</td><td>O</td><td>U</td><td>U</td><td colspan="2">——-</td></tr>
<tr><td>7</td><td>O</td><td>I</td><td>I</td><td>U</td><td>U</td><td>I</td><td>——-</td></tr>
<tr><td>8</td><td>O</td><td>O</td><td>O</td><td>U</td><td>O</td><td>O</td><td>O</td></tr>
</table>
<table>
<tr><td>Closeness Value</td><td>A</td><td>E</td><td>I</td><td>O</td><td>U</td><td>X</td></tr>
<tr><td></td><td>64</td><td>16</td><td>4</td><td>1</td><td>0</td><td>-1024</td></tr>
</table>
Minimum Department Preference (MDP) = I = 4, Sweep Width = 2, Number of iterations to be performed, N = 3, Assume Scale: 1 square = 400 m2, Length X Width for the layout: 10 X 24</td>
<td>10</td>
<td>L3</td>
<td>CO3</td>
</tr>
</table>

5.	Given the initial layout, flow matrix and cost matrix shown in figures 1,2 and 3, use the CRAFT scoring method and procedure two ways to obtain an improved layout.(consider any two possible exchanges).	10	L1	CO1

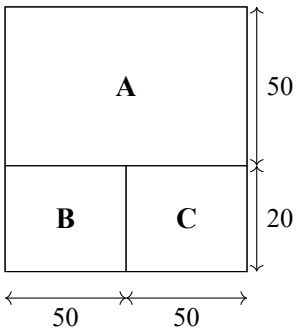


Fig 1. Initial Layout

From \ To	A	B	C
A		4	1
B	2		2
C	2	1	

Fig. 2. Flow Matrix

From \ To	A	B	C
A		1	1
B	1		1
C	1	1	

Fig. 3 Cost Matrix

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Date : 2nd May 2025	**CIE – 1**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Service Operations Management		**Course Code : IM365TDB**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is the primary goal of Service Operations Management?	2	1	1
2.	Name two major challenges faced by service operations managers in today's dynamic environment.	2	2	2
3.	Differentiate between B2B and B2C services with one example each.	2	3	1
4.	What type of service process is typically used in a fast-food restaurant, and why?	2	3	2
5.	How does service operations management differ from manufacturing operations management?	2	2	2
	Part – B			
1	Identify any 5 tools that have been used on you to collect information to enhance customer satisfaction. Now work backwards and assess what they were trying to understand and how they might have probably improved the service offering. (Use tabular columns and showcase structured assessment)	10	2	2
2	You are a Service Operations Manager of an online retail store. What challenges would you may face as the manager and what measures would you take to overcome them?	10	3	3
3	What are the different types of service processes? Cite the example of any service organization and highlight the different processes involved in that organization.	10	2	2
4	Explain the factors that you will assess as "Service Outcomes" for a service of your choice.	10	2	2
5	Describe a service from both an operational and customer perspective. Assess the mismatches between these perspectives.	5	4	3